

GLP-1 Receptor Agonists and Their Role in Tackling Obesity in Canada

In recent years, the prevalence of obesity in Canada has reached staggering levels. Data from 2015 suggests that around 27% of Canadian adults suffer from obesity, with around 35% of Canadian adults being overweight as well (Malla et al., 2023, p. 86). Numerous peer-reviewed studies have singled out obesity as a primary risk factor in the development of illnesses such as heart disease, cancer, diabetes, hypertension, and stroke among others. In addition, obesity has been identified as a burden that drives up health care expenditures (Malla et al., 2023, p. 83). As such, measures to combat the prevalence of obesity should be undertaken to prevent these negative health outcomes and societal costs (Malla et al., 2023, p. 91-92). The class of drugs known as GLP-1 (Glucagon-like peptide-1) receptor agonists has shown promise in aiding obesity management (Burcelin & Gourdy, 2016, p. 86). Thus, this paper will explore the function of GLP-1 receptor agonists (GLP-1 RAs) as they relate to obesity reduction, cover their proven efficacy in curbing obesity, and argue that this class of drugs provides Canadians both individual and societal benefits that outweigh any potential deleterious outcomes when used alongside other strategies such as a healthier diet and a more active lifestyle.

An Introduction to GLP-1 Receptor Agonists and Their Function as They Relate to Obesity Reduction

It is prudent to begin with exploring what exactly GLP-1 RAs are. The term “GLP-1” refers to “Glucagon-like peptide-1”. This is a hormone naturally found in the human body that is released into the bloodstream from the gut and pancreas after meals. Its release triggers signals that act upon the hypothalamus, limbic reward system, and the cerebral cortex, all of which are located in the brain. These signals induce feelings of fullness and the reduction of appetite and thus food consumption. They also act to slow the rate at which the stomach empties (Liuzzo &

Galiuto, 2021, p. 1652; Goldenberg et al., 2023, p. 2). According to the Irish Health Research Board's National Drugs Library (2023), an agonist is "a substance that mimics the actions of a neurotransmitter or hormone to produce a response when it binds to a specific receptor in the brain". So, substances in this class of drugs mimic the naturally occurring hormone GLP-1 and bind to GLP-1 receptors, thus bringing about an energy balance in the body that is beneficial for weight loss given the reduced consumption and cravings that follows (Wharton et al., 2023, p. 878; Goldenberg et al., 2023, p. 2). The implications for curtailing obesity are evident. If GLP-1 receptors are chronically activated, overall appetite and food consumption will invariably drop given the reduction of the underlying desire to engage in food intake, especially with the impact that GLP-1 receptor activation has on the brain's reward centres (Goldenberg et al., 2023, p. 2). It is interesting to note that the first explored use case for GLP-1 RAs was not the reduction of obesity. Rather, such drugs were first used to combat type 2 diabetes. Given the mutual causalities shared between type 2 diabetes and obesity, it would only be a matter of time before scientists observed the clinically significant weight reduction that GLP-1 RAs induced in clinical trials investigating their efficacy against type 2 diabetes (Brown et al., 2021, p. 266-267; Goldenberg et al., 2023, p. 1; Jastreboff et al., 2022, p. 206).

The Successful Track Record of GLP-1 Receptor Agonists in Obesity Reduction

A robust base of evidence exists to support the efficacy of GLP-1 RAs in obesity reduction and management. While there are several different chemical formulations in this class of drugs, a common theme is the significant advantage that they confer over weight-loss strategies that exclude them. Two notable examples covered here are semaglutide and orforglipron. Semaglutide is commonly marketed under the brand names Ozempic and Wegovy. In a 68-week long trial evaluating its impact in treating obesity, 1969 non-diabetic adult

participants with a BMI of at least 27 were at a 2:1 ratio administered a semaglutide or a placebo formulation. Both of these groups were put onto a reduced-calorie diet with a daily caloric deficit of 500 kcal and engaged in at least 150 minutes of physical activity per week. At the conclusion of the 68 weeks, the two groups achieved a mean weight loss of 14.9% and 2.4%, for the semaglutide and placebo-administered groups respectively. Those in the semaglutide-administered group were also observed to have improved in several areas associated with cardiometabolic health, such as blood pressure and glucose levels, as well as improved physical functioning compared to those in the placebo-administered group (Liuzzo & Galiuto, 2021, p. 1652).

Orforglipron, a promising recent formulation, has also shown remarkable efficacy in tackling obesity. This orally-delivered drug was developed as an answer to the barriers and challenges associated with injection-delivered drugs such as semaglutide. (Wharton et al., 2023, p. 878). During a trial with a 36-week treatment phase, non-diabetic adults participants with a BMI of at least 27 were split into various cohorts which were administered varying dosage strengths of orforglipron or a placebo formulation (Wharton et al., 2023, p. 878-879). The dosages administered to those receiving orforglipron were gradually increased as the trial progressed. Like the aforementioned semaglutide trial, this trial also incorporated a healthy eating and exercise component (Wharton et al., 2023, p. 879). Also as with the semaglutide trial, this trial yielded significant reductions in weight for those taking orforglipron as opposed to those in the placebo group. At week 36 of the trial, the mean change in weight ranged from a 9.4% reduction in those being administered 12 mg of orforglipron daily up to a 14.7% reduction in those being administered 45 mg daily. This was in contrast to a mere 2.3% reduction in the placebo group over the same period of time (Wharton et al., 2023, p. 880).

Addressing Observed Negative Side Effects

Studies on GLP-1 RAs have, in addition to demonstrating significant positive outcomes for patients, also observed some negative side effects of their usage. In the aforementioned study exploring the orally-administered drug orforglipron, a significant portion of the study population experienced adverse gastrointestinal effects, such as nausea, vomiting, and diarrhea. These adverse effects were more prevalent in those being administered orforglipron as opposed to the placebo treatment (Wharton et al., 2023, p. 881). However, most of these side effects were milder and moderate in severity. In addition to this, the side effects observed were mostly transient in nature and occurred midway through the increase in dosage levels as opposed to occurring at the highest dosage levels being administered. Notably, while a small number of study participants experienced more serious side effects, the incidence rates between those in the placebo and orforglipron-administered cohorts had no clinically-relevant difference, meaning the trial did not find any pertinent relationship between orforglipron and serious side effects (Wharton et al., 2023, p. 881). The prevalence of adverse gastrointestinal effects with orforglipron appeared to be a result of a dosage-escalation regime that would not be enforced in real-life usage of the drug. These findings of adverse effects occurring in particular stages of drug administration informed researchers about optimal and suboptimal dosage practices (Wharton et al., 2023, p. 887). Another area of concern that the study observed was pulse rate increases in those being administered orforglipron which were not similarly observed in those receiving the placebo treatment. Such pulse rate increases have been observed in other studies investigating GLP-1 RAs. However, as will be touched upon later, when taking the cardiovascular improvements that such drugs provide into account the negative potential of pulse rate increases is nullified (Wharton et al., 2023, p. 881, 887-888).

A similar set of results pertaining to drug safety arose in the 68-week long trial investigating semaglutide. In that trial, mild-to-moderate gastrointestinal effects were also observed, with such effects being transient in nature. Just under 10% of those being administered semaglutide experienced more serious side effects, compared to just over 6% of those being administered a placebo treatment. As with the study investigating orforglipron, negative side effects were associated with particular dosage escalation regimes, with modifications to dosage protocols mitigating said side effects (Liuzzo & Galiuto, 2021, p. 1652-1653). Long-term side effects of GLP-1 RA treatments are unclear, although a study of 304 participants being administered semaglutide that lasted two years found the same pattern of transient, mild-to-moderate gastrointestinal effects. In fact, this two-year study actually observed a greater percentage of those in the placebo group suffering serious side effects during the trial as opposed to those receiving semaglutide, with 11.8% of those in the placebo group and 7.9% of those in the semaglutide group experiencing these more serious adverse outcomes (Garvey et al., 2022, p. 2083, 2088).

Another concern that has been raised about GLP-1 RAs is a potential link between this class of drugs and suicidal ideation, and even self-harm. A potential link was first brought to the attention of the European Medicines Agency (EMA) (McIntyre et al., 2023, p. 47). An analysis of the available data suggested a disproportionate amount of reports pertaining to the GLP-1 RAs semaglutide and liraglutide, whereas the other drugs of this class that were investigated did not yield this disproportionality (McIntyre et al., 2023, p. 53). However, causation could not be established between these drugs and suicidal ideation. Furthermore, the study asserts that there is no: “plausible mechanism that would link exposure specifically to these two agents to suicidality that would not apply to the other agents within this class”. However, it appears that a more

plausible explanation here would be that those receiving treatment with these drugs are: “differentially affected by psychiatric disorders with known associated risk for suicide”. That is to say, given the unfortunate frequency of underlying mental illnesses and self-image issues in those eligible for GLP-1 RA treatment, this acts as a confounding factor in the relationship between these drugs and suicidal ideation (McIntyre et al., 2023, p. 54).

GLP-1 Receptor Agonists as A Public Good

Given the observed efficacy of GLP-1 RAs in tackling obesity, it is apparent that such drugs do have a place in Canada’s fight against obesity and its associated harms. Taking factors such as the loss of taxable income due to employment reduction, excess healthcare spending devoted to treating those with obesity-related illnesses, costs associated with obesity-related disabilities, and reduced tax revenue from a drop in overall consumption in those afflicted with obesity, one study assessed that the cost of obesity in 2021 alone on government and public resources in Canada was \$22.974 billion (Kotsopoulos & Connolly, 2024, p. 379). It further attributed 9.1% of all deaths in Canada to obesity, and in particular 10% of premature deaths of those between the ages of 20 and 64. This avoidable early loss of life, in addition to devastating families and communities, represents a substantial loss of public revenue and productive potential for our workforce (Kotsopoulos & Connolly, 2024, p. 387). The urgency of tackling the obesity crisis in Canada becomes more pressing when considering the expected 20%+ increase in the old-age dependency ratio. This ratio is between those 65 and older, who are generally no longer members of the broader workforce, and the portion of the population that comprises the bulk of the workforce, like those between the ages of 15 and 64. If left unchecked, obesity as it intersects with this demographic trend will lead to reduced output, lower tax revenue, greater

stress on public services and systems, and ultimately a lower standard of living for Canadians (Globerman, 2022; Kotsopoulos & Connolly, 2024, p. 387).

In order to combat obesity, a multi-faceted approach is needed. GLP-1 RAs should be incorporated into this strategy given their proven efficacy in combating and preventing obesity. When combined with commitment to a diet and regular physical exercise, reputable peer-reviewed studies have repeatedly demonstrated that the usage of these drugs is significantly more effective than dieting or engaging in exercise without taking such drugs (Liuzzo & Galiuto, 2021, p. 1652; Popoviciu et al., 2023, p. 11-12). That is not to say such drugs should be taken without other measures like dieting and exercising. Indeed, it has also been shown that GLP-1 RAs are less effective when taken without accompanying healthier diets and exercise regimens (Popoviciu et al., 2023, p. 12). To maximize their effectiveness and to help foster a healthier life in the long-term, they should be used where dieting and exercise alone have failed to mitigate and obesity. In such cases, a pharmacological supplement to aid weight loss is needed to bring about positive results (Burcelin & Gourdy, 2016, p. 87; Garvey et al., 2022, p. 2083).

While it is understandable to be skeptical of new pharmacological solutions to health issues such as obesity, especially given the lack of long-term data on potential side effects and the known short-term side effects, the societal consequences of obesity and its related outcomes far outweigh any downsides that GLP-1 RAs may pose. Short-term, transient gastrointestinal disorders pale in comparison in both severity and prevalence to the illnesses and conditions that obesity has been linked as a major risk factor for or linked to, such as heart disease, strokes, hypertension, diabetes, irregular pulmonary circulation, and even cancer (Malla et al., 2023, p. 83). As for concerns about potential long-term side effects one must consider the widespread usage of COVID-19 vaccines in an effort to mitigate the spread and severity of the COVID-19

pandemic. This measure was taken despite public and scientific debate and uncertainty regarding potential long-term side effects (Mandavilli, 2021; Suran, 2022, p. 23-24). If society is to avoid setting double standards for pharmacological solutions to public health crises it should thus embrace GLP-1 RAs as an effective tool in curbing its obesity crisis.

Conclusion

In closing, GLP-1 RAs are a proven aid to prevent and curb obesity, and should be adopted in Canada and beyond as part of a broader strategy to tackle this crisis. These drugs target root biological causes of obesity such as appetite and cravings for food. In this regard, they are highly effective, having been demonstrated in numerous peer-reviewed studies to have a significant edge over placebo treatments and dieting and physical exercise alone in spurring weight loss, as well as conferring positive cardiometabolic health outcomes and improving overall physical functioning. Despite skepticism of their usage, GLP-1 RAs provide these benefits without significant risks of serious adverse health outcomes, and any commonly observed negative side effects are relatively minor and transient in nature. There is simply a lack of plausible concerns about any long-term or more serious side effects occurring to any significant degree with the usage of these drugs. In addition to bringing about positive individual outcomes in mitigating obesity and related illness conditions, GLP-1 RAs are capable of providing profound societal benefits in reversing preventable, premature deaths and removing a substantial burden from public services and spending. Freeing up tens of billions of dollars annually yields numerous potential benefits for Canadian society, whether this means reducing the overall tax burden or committing this sum of money into other areas of concern, such as housing or more fruitful investments into the healthcare system. Mitigating obesity with the assistance of these drugs will also yield a more productive workforce, which will provide a

positive socioeconomic outcome for this country. In order to foster a holistic approach to weight reduction and a healthy lifestyle, GLP-1 RAs should be used in conjunction with a healthy diet and physical exercise to bring about long-term health benefits. Thus, GLP-1 RAs are a valuable asset to Canadian society, and provide far greater benefits to our society than any associated costs where used in a holistic, responsible manner.

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